

What is Databricks?

Databricks is a cloud-based unified data analytics platform designed to simplify and accelerate tasks like data engineering, data science, machine learning, and analytics workflows.

It is built on Apache Spark, and it integrates seamlessly with cloud storage and other data tools that helps to provide the collaborative environment for the teams to process, analyse, and visualize massive dataset efficiently.

Databricks in itself is a fully managed service and it is available on the clouds like AWS, Azure and Google Cloud.

In simpler terms, databricks provides us with a platform, where we can get the spark clusters and get the UI where we can write our pyspark code.

Key Features of Databricks File System (DBFS)

The key features of the databricks are:

- Unified platform: It combines data engineering, data science, and machine learning into one platform.
- It has Apache Spark integration (built on top of Apache spark which is very popular for distributed computing and data processing).
- It provides a collaborative environment (you can share your code like notebook, cluster or the computing environment itself).
- It provides scalability, (you can set an auto-scaling or you have a feature to scale your cluster based on your workloads).
- It provides seamless cloud integration (even if you have data available on AWS, ADLS or Google Cloud, you can bring the data in from that different cloud storage and still process and analyze).
- Delta Lake: It has the reliability and performance to the data lake using the delta lake capability, which provides asset transactions.
- Machine Learning
- Security and governance

Key Components

In Databricks we have a lot of components , some of the key component are:

1. **Workspace:** It is the collaborative environment for developers, where they can use this to develop and manage the data projects. Within the workspace itself we have notebooks, libraries, and dashboards.
2. **Cluster:** Clusters provide the computation power to run our logic. This computation power comes from the cluster, and these clusters are highly scalable meaning if you need two node clusters, you can use 2 nodes or if you need 200 node clusters, you can get that as well.
3. **Databricks notebooks:** It is an interactive way, where we can write our code in a multiple language and execute it.
4. **Jobs:** It is an automated and a scheduled task like ETL or running a pipeline or machine learning workflows.
5. **Databricks runtime:** It is an optimized version of Apache Spark with added features. Basically, the databricks has taken the spark and written some code on top of it and created its own environment known as the database's runtime.
6. **Delta Lake:** Databricks provides the asset transaction, through the delta lake, which is the default.
7. **Unity Catalog:** It provides the unity catalog, which can be used for doing data governance.

Use Cases

What could be a good use case of the databricks??

The databricks can be used in any of the organisation and projects which are looking for :

- Data Engineering Workloads (Building ETL pipelines).
- Data Science and Machine Learning
- Business Intelligence and Analytics (Insights and dashboards)
- Data Lakehouse (a modern data warehouse).
- Real-time analytics (stream processing).

Limitations

The following are some of the limitations of databricks:

- Cost
- Learning Curve
- Dependency on cloud providers
- Limited offline access.

Conclusion

Databricks is a versatile platform for managing your data workflows, and it provides the functionality in multiple use cases. In today's world, databricks is a go to choice for the organization who is going to develop their big data solutions.